



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 4 – Inner product spaces

Lecture 4.3 – Adjoint and QR factorisation

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Covectors

Lemma. Given a finite-dimensional inner product space $(V, \mathbb{F})$, for any linear transformation $T : V \rightarrow \mathbb{F}$ there is a **unique** vector $\mathbf{t} \in V$ such that $T(\mathbf{v}) = \langle \mathbf{t}, \mathbf{v} \rangle$ for all $\mathbf{v} \in V$.

Proof. Let $B = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be an orthonormal basis for V and $T : V \rightarrow \mathbb{F}$ be a linear map. We claim the vector $\mathbf{t} \in V$ for which $T(\mathbf{v}) = \langle \mathbf{t}, \mathbf{v} \rangle$ is

$$\mathbf{t} = \sum_{i=1}^n \overline{T(\mathbf{e}_i)} \mathbf{e}_i.$$

To confirm this, let $\mathbf{v} \in V$ with $[\mathbf{v}]_B = (\lambda_i)_{1 \leq i \leq n}$. Then

$$T(\mathbf{v}) = T\left(\sum_i \lambda_i \mathbf{e}_i\right) = \sum_i \lambda_i T(\mathbf{e}_i)$$

by linearity of T , and so

$$\langle \mathbf{t}, \mathbf{v} \rangle = \left\langle \sum_i \overline{T(\mathbf{e}_i)} \mathbf{e}_i, \sum_j \lambda_j \mathbf{e}_j \right\rangle = \sum_{i,j} \overline{T(\mathbf{e}_i)} \lambda_j \langle \mathbf{e}_i, \mathbf{e}_j \rangle = \sum_i \overline{T(\mathbf{e}_i)} \lambda_i = T(\mathbf{v}).$$

To see that this choice of $\mathbf{t}$ is unique, suppose there were some other $\mathbf{u} \in V$ for which $T(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v} \rangle$ for all $\mathbf{v} \in V$. Then $\langle \mathbf{t}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$ implying $\langle \mathbf{t} - \mathbf{u}, \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in V$, which in turn implies $\mathbf{t} - \mathbf{u} = \mathbf{0}$, so $\mathbf{u} = \mathbf{t}$.

Covectors

It follows that we can associate every linear map $T : V \rightarrow \mathbb{F}$ with a corresponding vector $\mathbf{t} \in V$. This motivates the following terminology.

Definition. Given a vector space $(V, \mathbb{F})$, any linear transformation from V to $\mathbb{F}$ is called a **covector**. The set of all such covectors is called the **dual space** of V , denoted $V^* = \mathcal{L}(V, \mathbb{F})$.

(Note that this definition doesn't assume existence of an inner product, and so holds for any scalar field $\mathbb{F}$, not just $\mathbb{R}$ or $\mathbb{C}$.)

Lemma. Given a finite-dimensional inner product space $(V, \mathbb{R})$, the mapping $\phi : V^* \rightarrow V$ given by $\phi(T) \rightarrow \mathbf{t}$ where $T(\mathbf{v}) = \langle \mathbf{t}, \mathbf{v} \rangle$ for all $\mathbf{v} \in V$ is an **isomorphism**.

Proof. Let $S, T \in V^*$ and $\lambda \in \mathbb{R}$. Then for all $\mathbf{v} \in V$ we have $(S + \lambda T)(\mathbf{v}) = S(\mathbf{v}) + \lambda T(\mathbf{v}) = \langle \phi(S), \mathbf{v} \rangle + \lambda \langle \phi(T), \mathbf{v} \rangle = \langle \phi(S) + \lambda \phi(T), \mathbf{v} \rangle$ so ϕ is linear. Furthermore ϕ is injective by the uniqueness of $\mathbf{t}$ for each T , and $\dim(V) = \dim(V^*)$, so ϕ is an isomorphism.

Over $\mathbb{C}$, we instead have a natural isomorphism between V and $\overline{V^*}$.

Definition. The mapping $T \mapsto \mathbf{t}$ is sometimes called **raising the index** and notated as $\mathbf{t} = T^\sharp$. The mapping $\mathbf{t} \mapsto T$ is sometimes called **lowering the index** and notated as $T = \mathbf{t}^\flat$.

Adjoint

Theorem. Given finite-dimensional inner product spaces V and W over $\mathbb{F}$, for any linear transformation $T : V \rightarrow W$ there is a **unique** corresponding linear transformation $T^* : W \rightarrow V$ for which $\langle \mathbf{w}, T(\mathbf{v}) \rangle = \langle T^*(\mathbf{w}), \mathbf{v} \rangle$ for all $\mathbf{v} \in V$ and $\mathbf{w} \in W$.

Proof. For each $\mathbf{w} \in W$, set $S_{\mathbf{w}} : V \rightarrow \mathbb{F}$ to be the map given by $S_{\mathbf{w}}(\mathbf{v}) = \langle \mathbf{w}, T(\mathbf{v}) \rangle$ for all $\mathbf{v} \in V$. Notice that $S_{\mathbf{w}}$ is linear since it is a composition of a linear map T and the operation of an inner product in the second argument.

Set $T^*(\mathbf{w}) = (S_{\mathbf{w}})^{\sharp}$, the vector $\mathbf{s} \in V$ for which $\langle \mathbf{s}, \mathbf{v} \rangle = S_{\mathbf{w}}(\mathbf{v})$ for all $\mathbf{v} \in V$. Then for each $\mathbf{w} \in W$ we have $\langle T^*(\mathbf{w}), \mathbf{v} \rangle = \langle \mathbf{s}, \mathbf{v} \rangle = S_{\mathbf{w}}(\mathbf{v}) = \langle \mathbf{w}, T(\mathbf{v}) \rangle$ for all $\mathbf{v} \in V$, as required.

Furthermore, this choice of T^* is unique, since if there were another map $S^* : W \rightarrow V$ for which $\langle \mathbf{w}, T(\mathbf{v}) \rangle = \langle S^*(\mathbf{w}), \mathbf{v} \rangle$, then we would have $\langle T^*(\mathbf{w}) - S^*(\mathbf{w}), \mathbf{v} \rangle = 0$ in which case $S^*(\mathbf{w}) = T^*(\mathbf{w})$ for all $\mathbf{w} \in W$ by positive definiteness.

Definition. Given finite-dimensional inner product spaces V and W over $\mathbb{F}$, for any linear transformation $T : V \rightarrow W$, the unique map T^* for which $\langle \mathbf{w}, T(\mathbf{v}) \rangle = \langle T^*(\mathbf{w}), \mathbf{v} \rangle$ for all $\mathbf{v} \in V$ and $\mathbf{w} \in W$ is called the **adjoint** of T .

Adjoint matrices

Example. Let $A \in M_{m,n}(\mathbb{C})$. Consider the induced linear map $T : \mathbb{C}^n \rightarrow \mathbb{C}^m$ given by $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{C}^n$. Then using the standard dot product as inner product, we have for all $\mathbf{w} \in \mathbb{C}^m$ that

$$\langle \mathbf{w}, T(\mathbf{v}) \rangle = \mathbf{w} \cdot (A\mathbf{v}) = \overline{\mathbf{w}}^T A\mathbf{v} = (A^T \overline{\mathbf{w}})^T \mathbf{v} = \overline{(\overline{A^T \mathbf{w}})}^T \mathbf{v} = (\overline{A^T \mathbf{w}}) \cdot \mathbf{v} = \langle \overline{A^T \mathbf{w}}, \mathbf{v} \rangle.$$

So the adjoint of T in this case is $T^* : \mathbb{C}^m \rightarrow \mathbb{C}^n$ given by $T^*(\mathbf{w}) = \overline{A^T \mathbf{w}}$ for all $\mathbf{w} \in \mathbb{C}^m$.

This justifies why we write $A^* = \overline{A^T}$ (note $T^*(\mathbf{w}) = A^* \mathbf{w}$) and why we call A^* the **adjoint** of the matrix A .

Theorem. Suppose $(V, \mathbb{F})$ and $(W, \mathbb{F})$ are finite-dimensional inner product spaces with **orthonormal** bases B and C respectively. Then for any linear transformation $T : V \rightarrow W$, writing $[T]_C^B = A$, we have $[T^*]_B^C = A^*$.

Proof. Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $C = \{\mathbf{c}_1, \dots, \mathbf{c}_m\}$. Then the (i, j) th entry of $A = [T]_C^B$ is the i th entry of $[T(\mathbf{b}_j)]_C$, meaning $A_{i,j} = \langle \mathbf{c}_i, T(\mathbf{b}_j) \rangle$ since C is orthonormal. Likewise, since B is orthonormal, the (i, j) th entry of $[T^*]_B^C$ is given by

$$\langle \mathbf{b}_i, T^*(\mathbf{c}_j) \rangle = \langle T(\mathbf{b}_i), \mathbf{c}_j \rangle = \overline{\langle \mathbf{c}_j, T(\mathbf{b}_i) \rangle} = \overline{A_{j,i}}.$$

So indeed $[T^*]_B^C = A^*$.

Properties of adjoints

For all finite-dimensional inner product spaces $(V, \mathbb{F})$ and $(W, \mathbb{F})$, the following properties hold.

Lemma. For all $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$, $(S + \lambda T)^* = S^* + \bar{\lambda} T^*$.

Proof. For all $\mathbf{v} \in V$ and $\mathbf{w} \in W$, we have

$$\begin{aligned}\langle (S + \lambda T)^*(\mathbf{w}), \mathbf{v} \rangle &= \langle \mathbf{w}, (S + \lambda T)(\mathbf{v}) \rangle = \langle \mathbf{w}, S(\mathbf{v}) \rangle + \bar{\lambda} \langle \mathbf{w}, T(\mathbf{v}) \rangle \\ &= \langle S^*(\mathbf{w}), \mathbf{v} \rangle + \bar{\lambda} \langle T^*(\mathbf{w}), \mathbf{v} \rangle \\ &= \langle (S^* + \lambda T^*)(\mathbf{w}), \mathbf{v} \rangle.\end{aligned}$$

So $(S + \lambda T)^* = S^* + \bar{\lambda} T^*$ by the uniqueness of adjoints.

Lemma. For all $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$, $(T \circ S)^* = S^* \circ T^*$.

Proof. For all $\mathbf{v} \in V$ and $\mathbf{w} \in W$, we have

$$\langle (T \circ S)^*(\mathbf{w}), \mathbf{v} \rangle = \langle \mathbf{w}, (T \circ S)(\mathbf{v}) \rangle = \langle T^*(\mathbf{w}), S(\mathbf{v}) \rangle = \langle (S^* \circ T^*)(\mathbf{w}), \mathbf{v} \rangle.$$

Lemma. For all $T \in \mathcal{L}(V, W)$, we have $(T^*)^* = T$.

Proof. For all $\mathbf{v} \in V$ and $\mathbf{w} \in W$, we have

$$\langle \mathbf{w}, (T^*)^*(\mathbf{v}) \rangle = \overline{\langle (T^*)^*(\mathbf{v}), \mathbf{w} \rangle} = \overline{\langle \mathbf{v}, T^*(\mathbf{w}) \rangle} = \langle T^*(\mathbf{w}), \mathbf{v} \rangle = \langle \mathbf{w}, T(\mathbf{v}) \rangle.$$

Unitary maps

Definition. A linear map $T : V \rightarrow V$ on a finite-dimensional inner product space $(V, \mathbb{F})$ is called **self-adjoint** (or **Hermitian**) if $T^* = T$.

Definition. An invertible linear map $T : V \rightarrow V$ on a finite-dimensional inner product space $(V, \mathbb{F})$ is called **unitary** if $T^* = T^{-1}$.

Lemma. A linear map T is unitary if and only if T^* is unitary.

Proof. Note $T^* = T^{-1} \iff \text{id} = T^* \circ T = T^* \circ (T^*)^* \iff (T^*)^* = (T^*)^{-1}$ so T is unitary if and only if T^* is unitary.

Definition. A linear map $T : V \rightarrow V$ on a finite-dimensional inner product space $(V, \mathbb{F})$ is called an **isometry** if $\|T(\mathbf{v})\| = \|\mathbf{v}\|$ for all $\mathbf{v} \in V$.

Lemma. A linear map is unitary if and only if it is an isometry.

Proof. For the forward direction, suppose T is unitary. Then $\|T(\mathbf{v})\|^2 = \langle T(\mathbf{v}), T(\mathbf{v}) \rangle = \langle T^*(T(\mathbf{v})), \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$, so it follows that $\|T(\mathbf{v})\| = \|\mathbf{v}\|$ and T is an isometry.

Now suppose T is an isometry. Then for all $\mathbf{v} \in V$ we have $0 = \|T(\mathbf{v})\|^2 - \|\mathbf{v}\|^2 = \langle T^*(T(\mathbf{v})) - \mathbf{v}, \mathbf{v} \rangle$, so we must have $T^*(T(\mathbf{v})) = \mathbf{v}$ for all $\mathbf{v} \in V$, meaning $T^* \circ T = \text{id}$ and so T is unitary.

Properties of unitary maps

Lemma. A linear map $T : V \rightarrow V$ is unitary if and only if $\langle T(\mathbf{v}), T(\mathbf{w}) \rangle = \langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{v}, \mathbf{w} \in V$.

Proof. For the forward direction, if T is unitary, then $\langle \mathbf{v}, \mathbf{w} \rangle = \langle T^*(T(\mathbf{v})), \mathbf{w} \rangle = \langle T(\mathbf{v}), T(\mathbf{w}) \rangle$ for all $\mathbf{v}, \mathbf{w} \in V$.

For the reverse direction, note in particular that $\langle \mathbf{v}, \mathbf{v} \rangle = \langle T(\mathbf{v}), T(\mathbf{v}) \rangle$ implying $\|\mathbf{v}\| = \|T(\mathbf{v})\|$ for all $\mathbf{v} \in V$. So T is an isometry and thus unitary.

Lemma. A linear map $T : V \rightarrow V$ is unitary if and only if whenever $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthonormal basis for V , so is $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)\}$.

Proof. For the forward direction, suppose T is unitary. Then from the previous lemma, we know $\langle T(\mathbf{e}_i), T(\mathbf{e}_j) \rangle = \langle \mathbf{e}_i, \mathbf{e}_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$, so $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)\}$ is an orthonormal basis.

For the reverse direction, suppose $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)\}$ are orthonormal bases and let $\mathbf{v} \in V$ so that $\mathbf{v} = \sum_i \langle \mathbf{e}_i, \mathbf{v} \rangle \mathbf{e}_i$. Then

$$\langle T(\mathbf{v}), T(\mathbf{v}) \rangle = \sum_i \sum_j \langle \mathbf{e}_i, \mathbf{v} \rangle \overline{\langle \mathbf{e}_j, \mathbf{v} \rangle} \langle T(\mathbf{e}_i), T(\mathbf{e}_j) \rangle = \sum_i |\langle \mathbf{e}_i, \mathbf{v} \rangle|^2 = \|\mathbf{v}\|^2,$$

implying $\|T(\mathbf{v})\| = \|\mathbf{v}\|$, and so T is an isometry and thus unitary.

Unitary matrices

Definition. A matrix $A \in M_{n,n}(\mathbb{C})$ is called:

- **symmetric** if and only if $A^T = A$.
- **orthogonal** if and only if $A^T = A^{-1}$.
- **self-adjoint** or **Hermitian** if and only if $A^* = A$.
- **unitary** if and only if $A^* = A^{-1}$.

Lemma. Given $A \in M_{n,n}(\mathbb{C})$, the columns of A form an **orthonormal basis** for $\mathbb{C}^n$ (with respect to the standard dot product) if and only if A is **unitary**.

Proof. Note A is unitary if and only if $A^*A = I$. But the (i, j) th entry of A^*A is just $\mathbf{v}_i^* \mathbf{v}_j = \langle \mathbf{v}_i, \mathbf{v}_j \rangle$ where $\mathbf{v}_k$ is the k th column of A , which by the right-hand side is 1 when $i = j$ and 0 otherwise. So the columns of A form an orthonormal basis if and only if A is unitary.

Corollary. The rows of a unitary matrix in $M_{n,n}(\mathbb{C})$ form an orthonormal basis for $\mathbb{C}^n$ (when transposed).

Example. Consider the matrix $A = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{i}{\sqrt{6}} & \frac{-i}{\sqrt{2}} \\ \frac{i}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ \frac{-i}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, which we showed in

Lecture 4.2 has an orthonormal basis of $\mathbb{C}^3$ as its columns. So A is unitary, and indeed $A^* = A^{-1}$. Also, the rows of A form an orthonormal basis of $\mathbb{C}^3$.

QR factorisation

Theorem. (QR factorisation)

Any matrix $A \in M_{m,n}(\mathbb{F})$ with $\text{rank}(A) = n \leq m$ can be written as $A = QR$ where $Q \in M_{m,n}(\mathbb{F})$ has orthonormal columns and $R \in \text{GL}(n, \mathbb{F})$ is upper triangular (or “right triangular”).

Proof. Recall the Gram-Schmidt process produces an orthonormal basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ from a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ by constructing new vectors

$$\mathbf{w}_k = \mathbf{v}_k - \sum_{i=1}^{k-1} \text{proj}_{\mathbf{w}_i}(\mathbf{v}_k) \text{ and } \mathbf{e}_k = \frac{\mathbf{w}_k}{\|\mathbf{w}_k\|} \text{ for each } 1 \leq k \leq n$$
$$\implies \mathbf{v}_k = \mathbf{w}_k + \sum_{i=1}^{k-1} \frac{\langle \mathbf{w}_i, \mathbf{v}_k \rangle}{\|\mathbf{w}_i\|^2} \mathbf{w}_i = \|\mathbf{w}_k\| \mathbf{e}_k + \sum_{i=1}^{k-1} \langle \mathbf{e}_i, \mathbf{v}_k \rangle \mathbf{e}_i.$$

Since A has rank equal to its number of columns, its columns form a basis for $\text{im}(A)$, so Gram-Schmidt gives

$$A = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_n) = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \dots \quad \mathbf{e}_n) \begin{pmatrix} \|\mathbf{w}_1\| & \langle \mathbf{e}_1, \mathbf{v}_2 \rangle & \dots & \langle \mathbf{e}_1, \mathbf{v}_n \rangle \\ 0 & \|\mathbf{w}_2\| & \dots & \langle \mathbf{e}_2, \mathbf{v}_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \|\mathbf{w}_n\| \end{pmatrix}$$

where the first matrix has orthonormal columns and the second is upper triangular (and invertible since it has nonzero determinant).

Example

Exercise. Find a QR factorisation of $M = \begin{pmatrix} 1 & 0 & 0 \\ i & 1 & 0 \\ -i & 0 & 1 \end{pmatrix}$.

Solution. From the Gram-Schmidt example in Lecture 4.2, we can find

$$M = QR \text{ where } Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{i}{\sqrt{6}} & \frac{-i}{\sqrt{2}} \\ \frac{i}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ \frac{-i}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix} \text{ and } R = \begin{pmatrix} \sqrt{3} & \frac{-i}{\sqrt{3}} & \frac{i}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

(Note that we could read off each $\langle \mathbf{e}_i, \mathbf{v}_j \rangle$ term by taking the scalar part of $\text{proj}_{\mathbf{w}_i}(\mathbf{v}_j)$ and taking the square root of its denominator.)

Exercise. Find a QR factorisation of $A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 1 \end{pmatrix}$.

Solution. First apply Gram-Schmidt to the two column vectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

$$\text{Then } \mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \mathbf{w}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{w}_1}(\mathbf{v}_2) = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{(-1)}{5} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -4 \\ 2 \\ 5 \end{pmatrix}.$$

$$\text{So } A = QR \text{ where } Q = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{-4}{3\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{2}{3\sqrt{5}} \\ 0 & \frac{5}{3\sqrt{5}} \end{pmatrix} \text{ and } R = \begin{pmatrix} \sqrt{5} & \frac{-1}{\sqrt{5}} \\ 0 & \frac{3\sqrt{5}}{5} \end{pmatrix}.$$

Square QR factorisation

Notice that when a matrix A is square, the Q matrix in the QR factorisation of A is also square, and therefore unitary.

If $A \in M_{m,n}(\mathbb{F})$ is not square with $\text{rank}(A) = n < m$, we could find an alternative “square” QR factorisation of A by appending columns to A until it is square with rank m , finding the corresponding QR factorisation normally, and appending $m - n$ zero rows to R .

Exercise. Find a QR factorisation of $A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 1 \end{pmatrix}$ where Q is unitary.

Solution. We apply Gram-Schmidt as before, but this time with an extra third vector. If we can easily choose the vector to be orthogonal to the columns of A , the projections in the Gram-Schmidt calculation will just be 0.

In this case we can see $\mathbf{v}_3 = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ will be orthogonal to each column.

So $A = QR$ where $Q = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{-4}{3\sqrt{5}} & \frac{2}{3} \\ \frac{2}{\sqrt{5}} & \frac{2}{3\sqrt{5}} & \frac{-1}{3} \\ 0 & \frac{5}{3\sqrt{5}} & \frac{2}{3} \end{pmatrix}$ and $R = \begin{pmatrix} \sqrt{5} & \frac{-1}{\sqrt{5}} \\ 0 & \frac{3\sqrt{5}}{5} \\ 0 & 0 \end{pmatrix}$.

Projection minimises distance

One area where QR factorisations prove useful is the [method of least squares](#). Suppose we have a system of linear equations for which there are more equations than there are unknown variables, but no exact simultaneous solution (e.g. sampled data points). If we want to find the “best” solution, we are seeking the solution that is “closest” to the collection of data points. We can invoke a notion of “closest” with respect to distance as interpreted through an inner product.

In general, projection of a vector $\mathbf{v}$ onto a subspace W produces the closest vector to $\mathbf{v}$ within W :

Lemma. Given a finite-dimensional inner product space V with $\mathbf{v} \in V$ and a subspace $W \leq V$, for every $\mathbf{w} \in W$ we have $\|\mathbf{v} - \mathbf{w}\| \geq \|\mathbf{v} - \text{proj}_W(\mathbf{v})\|$ with equality if and only if $\mathbf{v} - \mathbf{w} \in W^\perp$ (meaning $\mathbf{w} = \text{proj}_W(\mathbf{v})$).

Proof. Let $\mathbf{v} \in V$ and $\mathbf{w} \in W \leq V$, and set $\mathbf{p} = \text{proj}_W(\mathbf{v})$. Then

$$\|\mathbf{v} - \mathbf{w}\|^2 = \|(\mathbf{v} - \mathbf{p}) - (\mathbf{w} - \mathbf{p})\|^2 = \|\mathbf{v} - \mathbf{p}\|^2 + \|\mathbf{w} - \mathbf{p}\|^2 \geq \|\mathbf{v} - \mathbf{p}\|^2,$$

where in the middle step we have used the fact $\mathbf{v} - \mathbf{p}$ and $\mathbf{w} - \mathbf{p}$ are orthogonal. Note equality holds only when $\|\mathbf{w} - \mathbf{p}\| = 0$, implying $\mathbf{w} = \mathbf{p}$ by positive definiteness, and so $\mathbf{v} - \mathbf{w} \in W^\perp$.

Method of least squares

In the situation that we wish to find a “closest” solution to a set of data points in the form of a system of linear equations $A\mathbf{x} = \mathbf{b}$, we wish to minimise the value of $\|A\mathbf{x} - \mathbf{b}\|$.

Definition. Given a system of equations $A\mathbf{x} = \mathbf{b}$ for some $A \in M_{m,n}(\mathbb{F})$, $\mathbf{x} \in \mathbb{F}^n$ and $\mathbf{b} \in \mathbb{F}^m$, the solutions to the problem of minimising $\|A\mathbf{x} - \mathbf{b}\|$ are called the **least squares solutions** for the system.

Theorem. Given a system of equations $A\mathbf{x} = \mathbf{b}$, a least squares solution to the system is a solution to the system of equations $A^*A\mathbf{x} = A^*\mathbf{b}$.

Proof. Per the previous lemma, writing $W = \{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \text{col}(A)$, we have that $\|A\mathbf{x} - \mathbf{b}\|$ will be minimised when $A\mathbf{x} - \mathbf{b} \in W^\perp$. So we want to find $\mathbf{x} \in \mathbb{F}^n$ such that $\langle A\mathbf{x} - \mathbf{b}, A\mathbf{v} \rangle = 0$ for all $\mathbf{v} \in \mathbb{F}^n$. Invoking the adjoint, we have $\langle A^*(A\mathbf{x} - \mathbf{b}), \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in \mathbb{F}^n$, and so we must have $A^*A\mathbf{x} - A^*\mathbf{b} = \mathbf{0}$ meaning $A^*A\mathbf{x} = A^*\mathbf{b}$.

Definition. Given a system of equations $A\mathbf{x} = \mathbf{b}$ as above, we call the system of equations $A^*A\mathbf{x} = A^*\mathbf{b}$ its corresponding **normal equations**.

Fact. The system of normal equations always has a solution, and if the columns of A are **linearly independent**, then the solution is unique and given by $\mathbf{x} = (A^*A)^{-1}A^*\mathbf{b}$.

Example – Line of best fit

Exercise. Find the least squares line of best fit to the points $(0, 1), (1, 3), (2, 2), (3, 0)$.

Solution. If there were an exact solution, we would want to find $c, m \in \mathbb{R}$ for which $y_i = c + mx_i$ for each pair of points (x_i, y_i) . So we want the “closest” (i.e. least squares) solution $\mathbf{x} = \begin{pmatrix} c \\ m \end{pmatrix}$ to $A\mathbf{x} = \mathbf{b}$ where

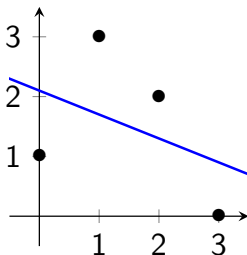
$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 0 \end{pmatrix}. \text{ The associated normal equations } A^*A\mathbf{x} = A^*\mathbf{b}$$

$$\text{give } \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 6 \\ 7 \end{pmatrix}, \text{ from which we get the solution } \mathbf{x} = \frac{1}{10} \begin{pmatrix} 21 \\ -4 \end{pmatrix}.$$

So the line of best fit is $y = \frac{21}{10} - \frac{2}{5}x$.

Example – Line of best fit

We can plot the result to check this:



In general, when asked to find the least squares line of best fit for n sample points $(x_i, y_i) \in \mathbb{R} \times \mathbb{R}$, we will always have $\mathbf{b} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$, $A = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}$, and

$$A^*A = A^T A = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}.$$

Using QR factorisations

Lemma. If a matrix $A \in M_{m,n}(\mathbb{R})$ has a QR factorisation $A = QR$, the least squares solution to $A\mathbf{x} = \mathbf{b}$ is given by $\mathbf{x} = R^{-1}Q^*\mathbf{b}$.

Proof. Given $A = QR$, the normal equations $A^*A\mathbf{x} = A^*\mathbf{b}$ can be written as $R^*Q^*QR\mathbf{x} = R^*Q^*\mathbf{b}$. Since the columns of Q are orthonormal, we know $Q^*Q = I$ and so in fact $R^*R\mathbf{x} = R^*Q^*\mathbf{b}$. Since R (and so also R^*) is invertible, we can simplify this further to $\mathbf{x} = R^{-1}Q^*\mathbf{b}$.

Exercise. Use QR factorisation to find the least squares line of best fit to the points $(0, 1), (1, 3), (2, 2), (3, 0)$.

Solution. We want the least squares solution $\mathbf{x} = \begin{pmatrix} c \\ m \end{pmatrix}$ to $A\mathbf{x} = \mathbf{b}$ where

$$\mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 0 \end{pmatrix} \text{ and } A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{3}{2\sqrt{5}} \\ \frac{1}{2} & -\frac{1}{2\sqrt{5}} \\ \frac{1}{2} & \frac{1}{2\sqrt{5}} \\ \frac{1}{2} & \frac{3}{2\sqrt{5}} \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 0 & \sqrt{5} \end{pmatrix} \text{ as a QR}$$

factorisation. So $\mathbf{x} = R^{-1}Q^*\mathbf{b} = \frac{1}{10} \begin{pmatrix} 21 \\ -4 \end{pmatrix}$ and we again find the line of best fit is $y = \frac{21}{10} - \frac{2}{5}x$.

Example – Quadratic of best fit

Exercise. Find the least squares quadratic of best fit to the points $(0, 1), (1, 3), (2, 2), (3, 0)$.

Solution. If there were an exact solution, we would want to find $a_0, a_1, a_2 \in \mathbb{R}$ for which $y_i = a_0 + a_1x_i + a_2x_i^2$ for each pair of points (x_i, y_i) .

Here we want the least squares solution $\mathbf{x} = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$ to $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 0 \end{pmatrix}. \text{ The associated normal equations}$$

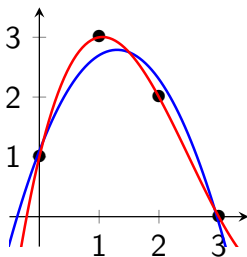
$$A^*A\mathbf{x} = A^*\mathbf{b} \text{ give } \begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 6 \\ 7 \\ 11 \end{pmatrix}, \text{ from which we can get the}$$

$$\text{solution } \mathbf{x} = \frac{1}{10} \begin{pmatrix} 11 \\ 26 \\ -10 \end{pmatrix}.$$

So the line of best fit is $y = \frac{11}{10} + \frac{13}{5}x - x^2$.

Example – Quadratic of best fit

We can again plot the result (in blue) to check this:



We could repeat the exercise for the least squares cubic of best fit. In this case, since there are only four sample points, there is a unique cubic that passes through them all, and the least squares method will find it. (Exercise: Prove this!)

Exercise. Show that the cubic through the points $(0, 1)$, $(1, 3)$, $(2, 2)$, $(3, 0)$ is given by $y = 1 + \frac{25}{6}x - \frac{5}{2}x^2 + \frac{1}{3}x^3$.